

Day 9 — Cross-cultural Lunar Calendars + Project Build

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min (15 min content + 30 min project)

Learning objective

Students will compare the Indian lunisolar calendar with three other lunar/lunisolar calendars (Islamic Hijri, Chinese, Hebrew) — identifying what is shared (lunar month basis) and what differs (intercalation rules).

Materials

- Board with a comparison table ready
- Project supplies (continued from Day 8)
- Notebooks

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*Many cultures, looking at the same Moon, built different calendars. Today: a 15-minute compare-and-contrast, then back to project build.*”

Core content (15 min)

1. **The common starting point.** Every culture observing the Moon notices the ~29.5-day cycle. The question every culture answers: *how do we fit 29.5-day lunar months into a 365.24-day solar year?*

The three resolutions:

- **Ignore the solar year.** Use only lunar months. The months drift through the seasons. (Islamic / Hijri)
 - **Re-align periodically with an extra month.** Lunisolar. (Indian, Chinese, Hebrew)
 - **Ignore the lunar month.** Use only the solar year, fix months at 30–31 days. (Modern Gregorian)
2. **The four calendar table.** Build on the board:

Feature	Indian (Hindu)	Islamic (Hijri)	Chinese	Hebrew
Type	Lunisolar	Pure lunar	Lunisolar	Lunisolar
Months/ year	12 (or 13 with <i>adhika māsa</i>)	12	12 (or 13 with leap month)	12 (or 13 with <i>Adar II</i>)
New	<i>amāvasyā</i> or <i>pūrṇimā</i>	First sighting of crescent	<i>amāvasyā</i>	<i>amāvasyā</i>

Feature	Indian (Hindu)	Islamic (Hijri)	Chinese	Hebrew
month begins at	(region-dependent)	after new moon	(new moon)	(with calculated rules)
Days per year (average)	~365.25	~354	~365.25	~365.25
Intercalation rule	Month with no <i>saṃkrānti</i> becomes <i>adhika</i> (extra)	None	Month with no <i>zhōngqì</i> (zhongqi solar term)	7 leap months in 19 years (Metonic)
Drift through seasons	Stays aligned	Drifts ~11 days/year backward	Stays aligned	Stays aligned

3. **A specific example.** *Ramaḍān* (Islamic) shifts ~11 days earlier each Gregorian year. *Caitra śukla pratipadā* (Indian new year, *Ugādi/Gudi Pādvā*) stays in March–April. *Pesach* (Hebrew) stays in spring. *Chūn Jié* (Chinese new year) stays in late January / early February. The lunisolar systems all use intercalation to keep their months in-season; the pure lunar system does not.

4. **Why does any of this matter?** Three reasons:

- **Cultural literacy.** Knowing how other calendars work helps in a multi-cultural society.
- **Astronomical insight.** Each calendar reveals a different choice in handling the same underlying mismatch (29.5 vs. 365.24).
- **Historical depth.** All four systems were refined over many centuries. The Indian, Chinese, and Hebrew systems independently arrived at lunisolar intercalation — convergent astronomical reasoning.

Activity (25 min) — Project build

Pairs continue their projects from Day 8. The teacher rotates with two priorities:

1. **Source check.** Every project has a real, named citation by end of class today. No invented references.
2. **3-minute script check.** Every pair has written or rehearsed a 3-minute presentation script. The artefact alone is not the project; the *explanation* is.

Pairs whose project relates to cross-cultural lunar calendars can use the Day 9 content directly. Others are encouraged to add a one-sentence cross-cultural note (“this Indian system, like the Chinese and Hebrew, is lunisolar…”).

Wrap-up (5 min)

- All pairs submit a near-final draft (photo / file).

- One sentence from each pair on the board: **“Our project covers ____ and our source is ____.”**
- **Tomorrow:** Day 10. 3-minute presentations. Summative quiz.

Student-facing content

Four lunar/lunisolar calendars compared:

- **Indian (lunisolar)** — 12 months, *adhika māsa* every ~32 months.
- **Islamic / Hijri (pure lunar)** — 12 months, no intercalation. Drifts ~11 days/year.
- **Chinese (lunisolar)** — 12 months, leap month every ~32 months.
- **Hebrew (lunisolar)** — 7 leap months in 19 years (Metonic cycle).

The Indian, Chinese, and Hebrew systems independently arrived at lunisolar intercalation. Their rules differ; the underlying problem is the same.

Homework

- Continue the Moon diary (one night left).
- Final touches on your project. Bring everything to class tomorrow.

Differentiation

- **One tier up:** Pick TWO calendars and write a paragraph: under what circumstances does each calendar’s design choice make sense? (Pure lunar — when religious months not tied to season matter most; lunisolar — when agriculture matters; pure solar — when civil administration matters.)
- **One tier down:** Compare just the Indian and the Hijri (pure lunar). One difference, one similarity. That’s enough.

Teacher notes

- This content is genuinely cross-cultural. Avoid framing as “ours is best.” All four systems work; they encode different priorities.
- The Metonic cycle (19-year Hebrew cycle) and the Indian intercalation rule converge to about the same accuracy. This is mathematically interesting — students at Senior band can explore it.
- Keep this part to 15 minutes. The project is the major work today.

Citation(s) used in this lesson

- General references for each calendar — use any standard encyclopaedic source (e.g. *Encyclopaedia Britannica* entries on the Hijri, Chinese, and Hebrew calendars). Do not invent specific verse numbers from primary sources.